Reminders

<u>Upcoming due dates</u>

Wed Assignment 1
    Project review (1 per group)
Fri Discussion lab 3

Repo invites! Click accept before it expires!

Section this week covers data visualization

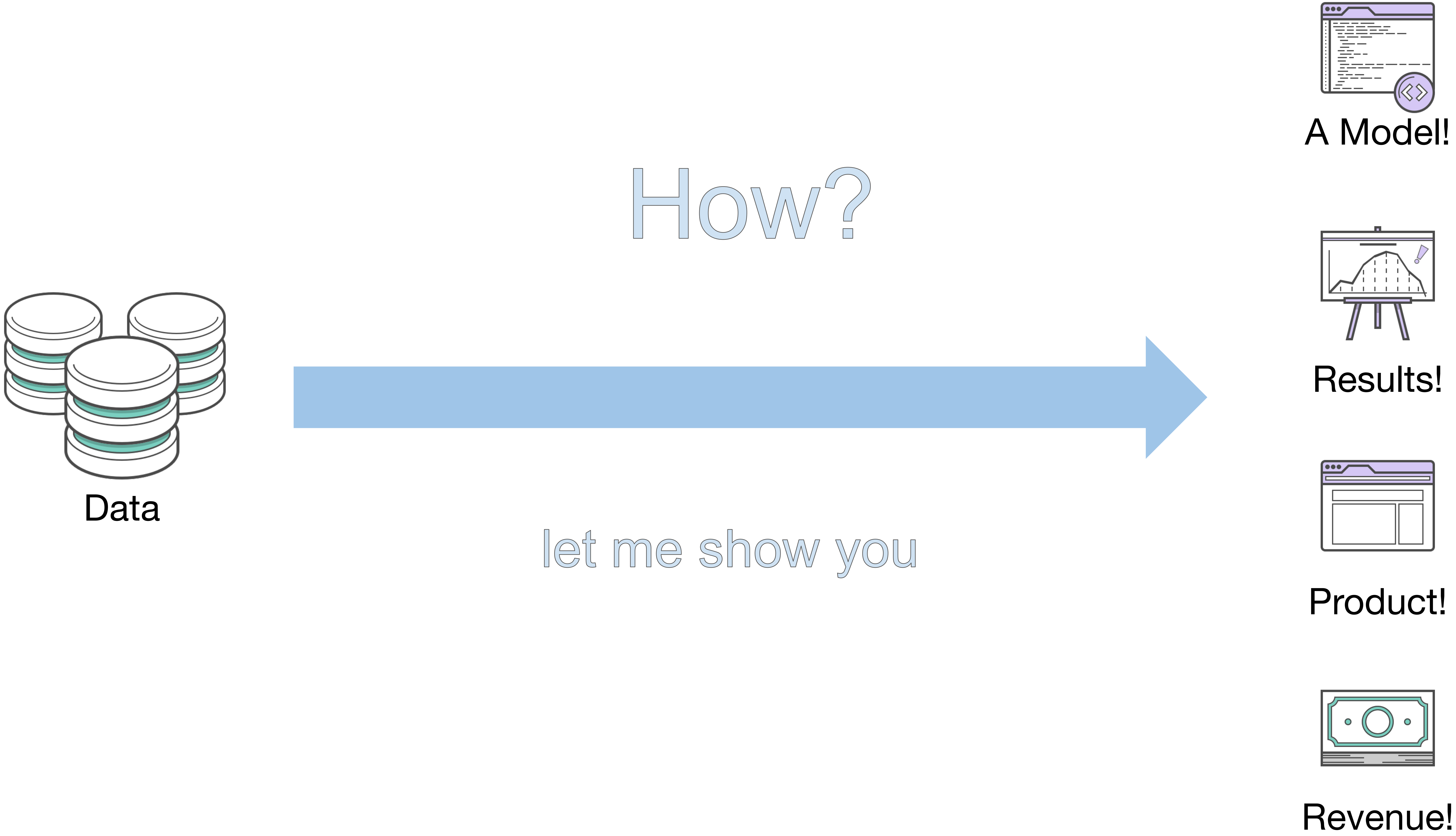
# Analysis toolbox

## Data Science in Practice

Jason G. Fleischer, PhD    Dept. of Cognitive Science    UC San Diego    https://jgfleischer.com


THE BASICS OF
TECHNICAL ANALYSIS
ALLEGRO
DECLINE
PROLOGUE
DOLDRUMS
SPLINE
HMM!
LUMBAR SUPPORT
RENEWAL
HARK! THE CLIFFS!
THESE TWO POINTS DEFINE A LINE! PROMISING SIGNAL.
DECLINATION
BATHTUB
UPTALK
INFLECTION
YIKES
WRONG!
COULD BE AN OMEN?
SLOPE!
LIKELY TO CONTINUE FOREVER
RED+GREEN =CHRISTMAS!
IF I ADD SOME LINES HERE I CAN CONVINCE MYSELF I'M DOING SOMETHING MORE THAN JUST SEEING PATTERNS IN A GRAPH OF A RANDOM WALK

xkcd.com


How?
let me show you
Data
A Model!
Results!
Product!
Revenue!

Data
The Analytic Approach
Your Tool Box
The Goods
Descriptive Analysis
Exploratory Analysis
?
Inferential Analysis
Predictive Analysis
Causal Analysis
Mechanistic Analysis

Data
The Analytic Approach
Your Tool Box
The Goods
Descriptive Analysis
Exploratory Analysis
?
Inferential Analysis
Predictive Analysis
Causal Analysis
Mechanistic Analysis
Classic Statistics (parametric & nonparametric)
Frequentist & Bayesian
Text & Geospatial Analysis
Statistical learning/ML
- Supervised
- Unsupervised
Monte Carlo simulations
variable X
causes
variable Y
e.x. effects of new medication on some illness by randomized trial
variable X 3.2 units
results in
variable Y 1.1 units
e.x. electric current governed by wire size

# Summary: Analytical Approaches

*Typically Less Effort*

Model making

## Descriptive Analysis

- 1st thing you do on new data
- Summarize the data
- univariate plots of variables

## Exploratory Analysis

- Exploring relationships
- Asking/definning questions
- univariate/bivariate/multivariate analysis and plotting
- formulate hypothesis

## Inferential Analysis

- Estimating uncertainty
- test theories (infer) about the population (data gen. process)
- Building inference models

Model making

Model making

Model making

*Typically More Effort*

## Predictive Analysis

- Building predictive models
- Use historical knowledge to predict future events
- Finding patterns

## Mechanistic Analysis

- Understand precise changes one variable has on another
- typically modeled using deterministic equations
- break down complex systems into constituent parts

## Causal Analysis

- Determine the average change in one variable when you alter another
- typically requires experiments (e.g. randomized studies)
- manipulate one variable observe effect on other

General question: What impacts politics in America?

Data Science question: Is there a relationship between the sentiment of political words in South Park and America's presidential approval rating?

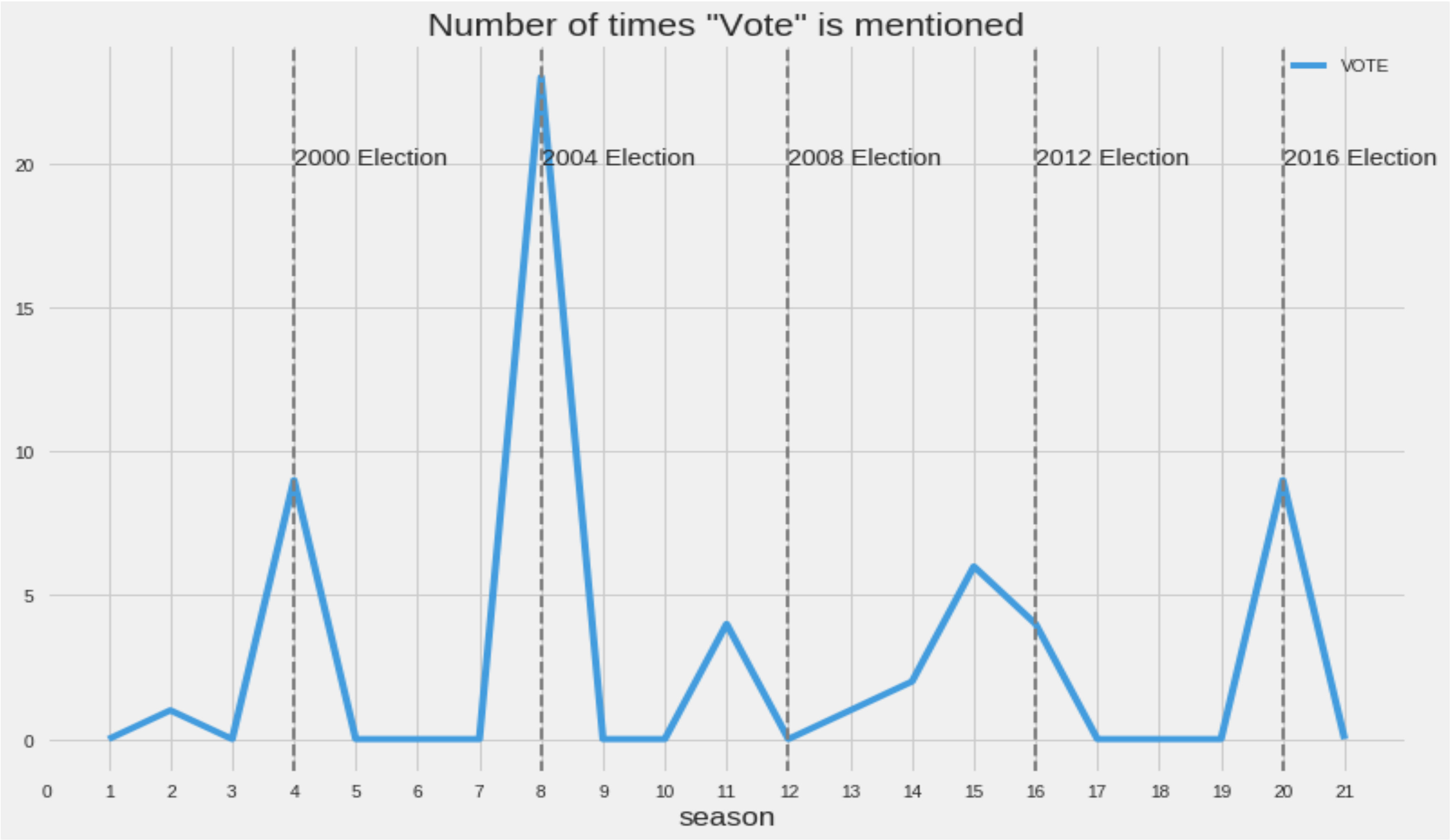
Descriptive

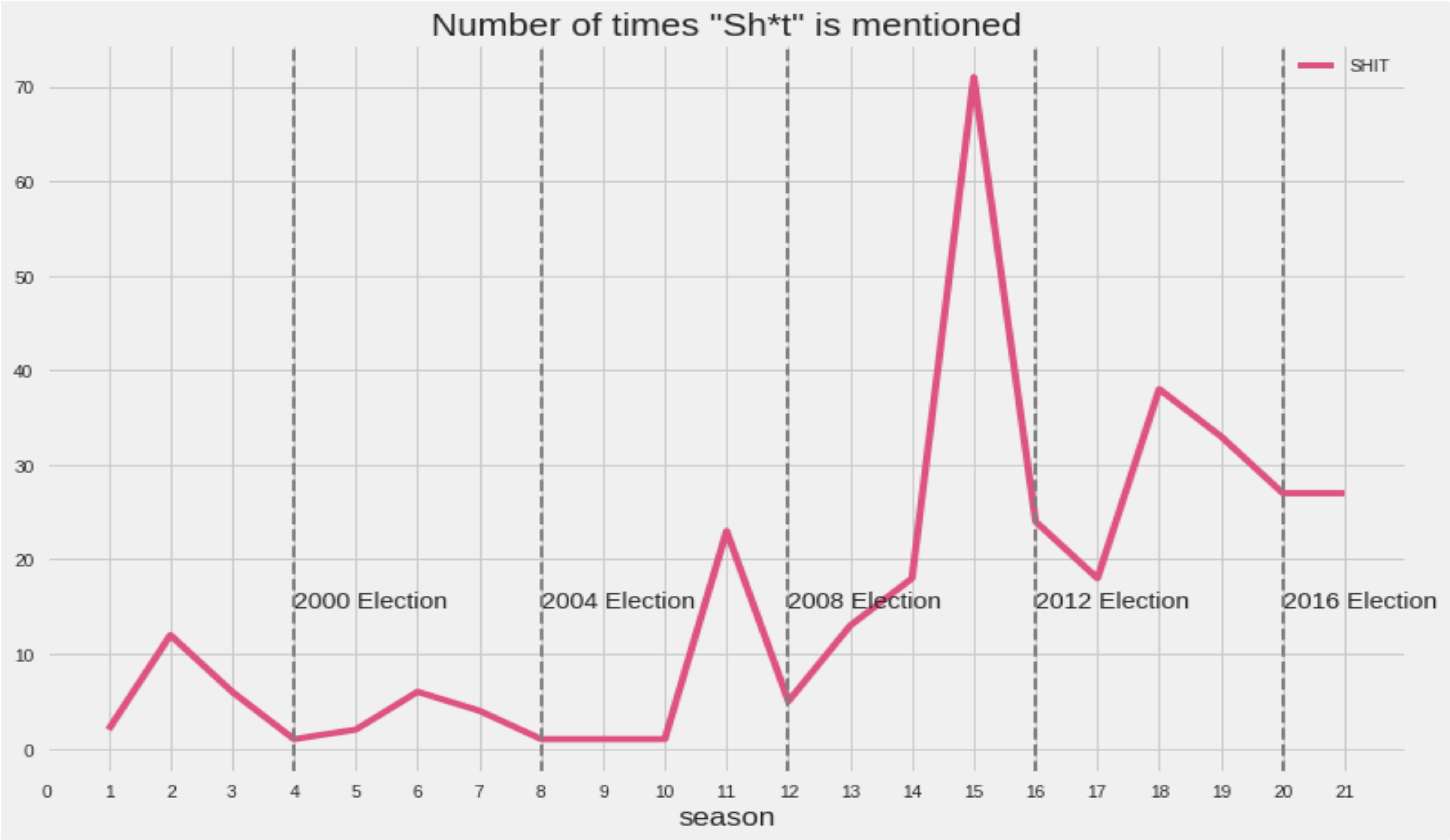
Exploratory

Inferential

Text Analysis

Classic Statistics (parametric & nonparametric)




https://github.com/COGS108/FinalProjects-Wi18/blob/master/071-FinalProject.ipynb

General question: How has COVID-19 impacted students?

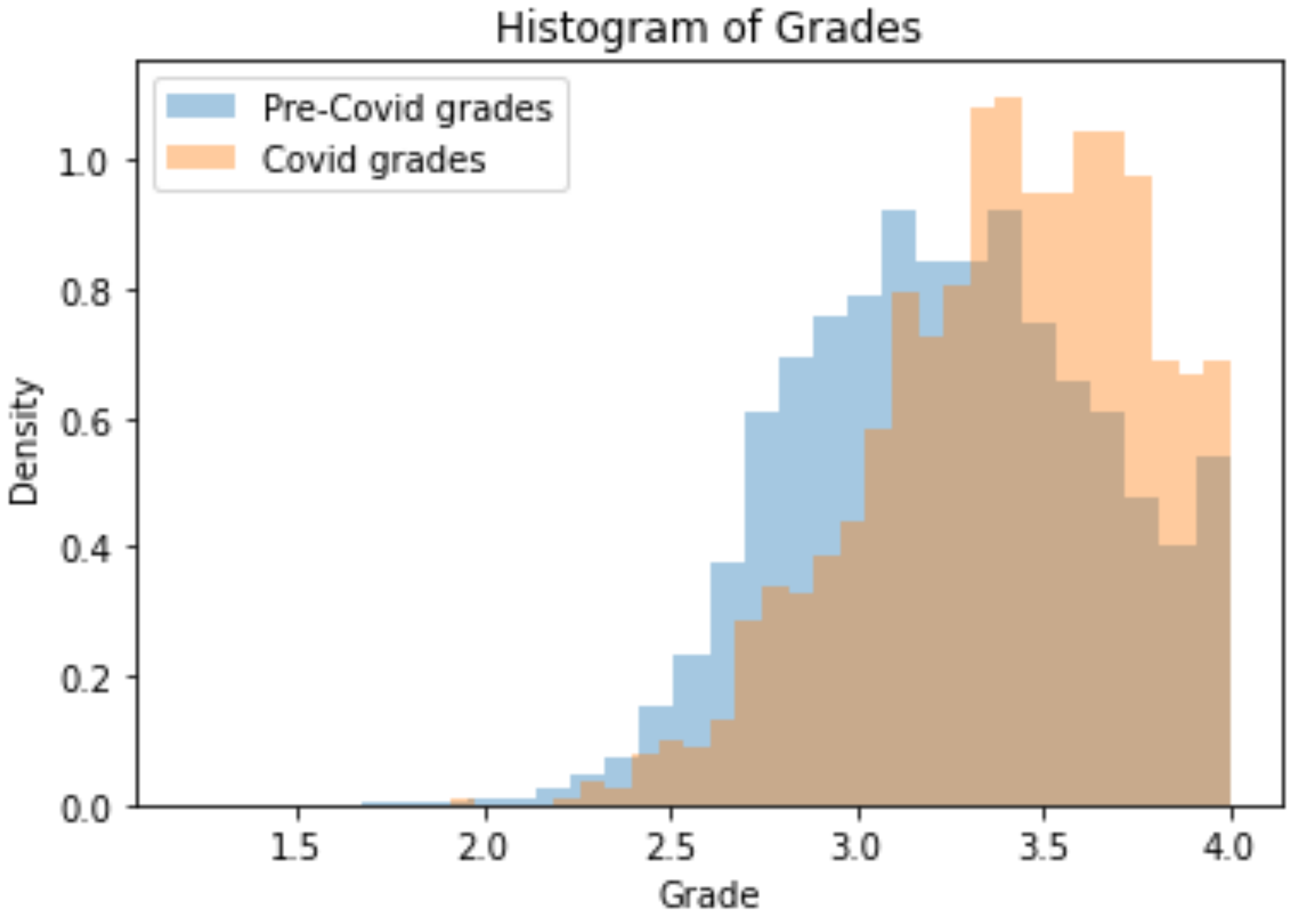
Data Science question: At UCSD, is there a difference between students' grades and how they rate their classes before COVID-19 and during remote learning, due to COVID-19?

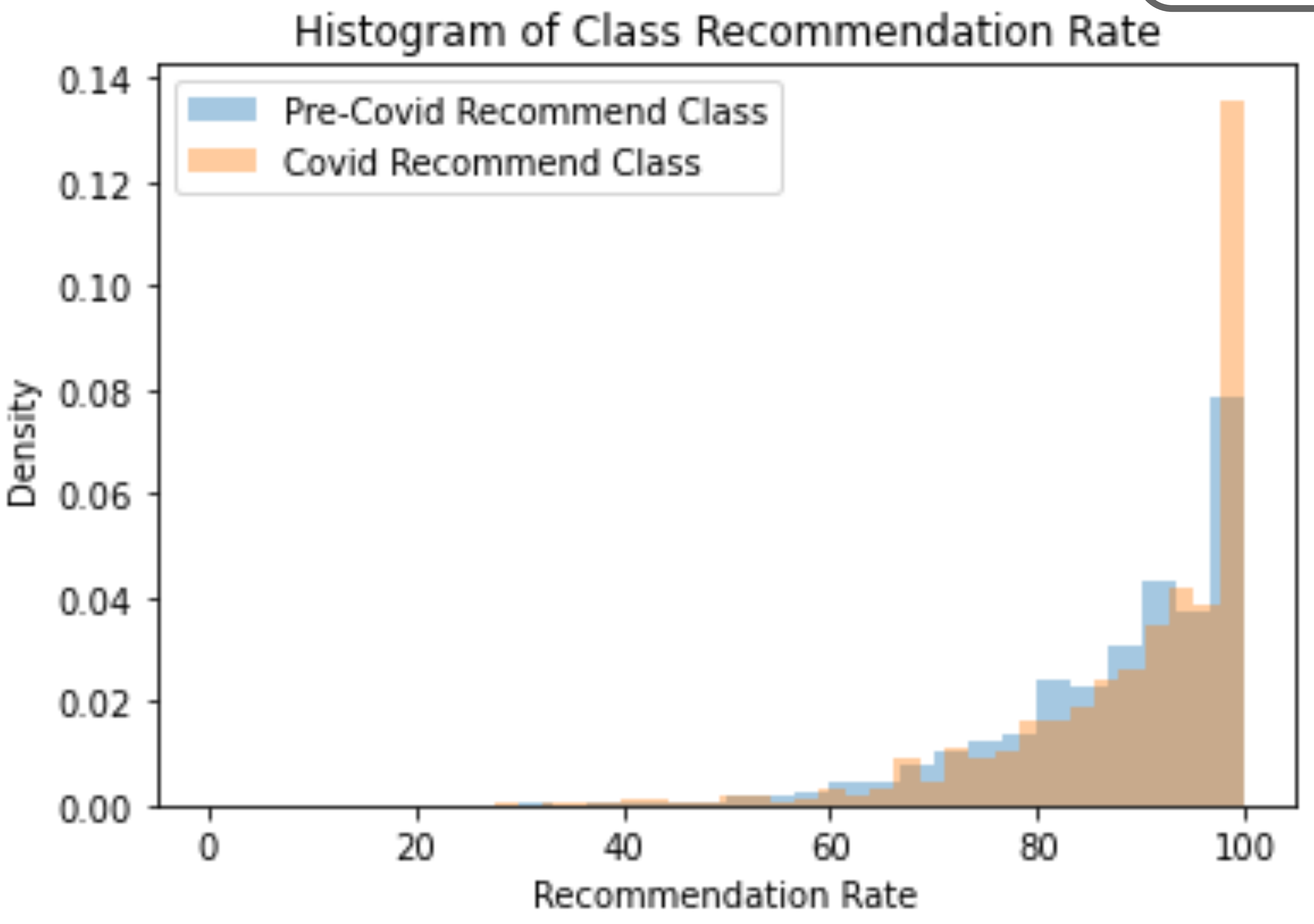
**Descriptive**

**Exploratory**

**Inferential**

Classic Statistics (parametric & nonparametric)

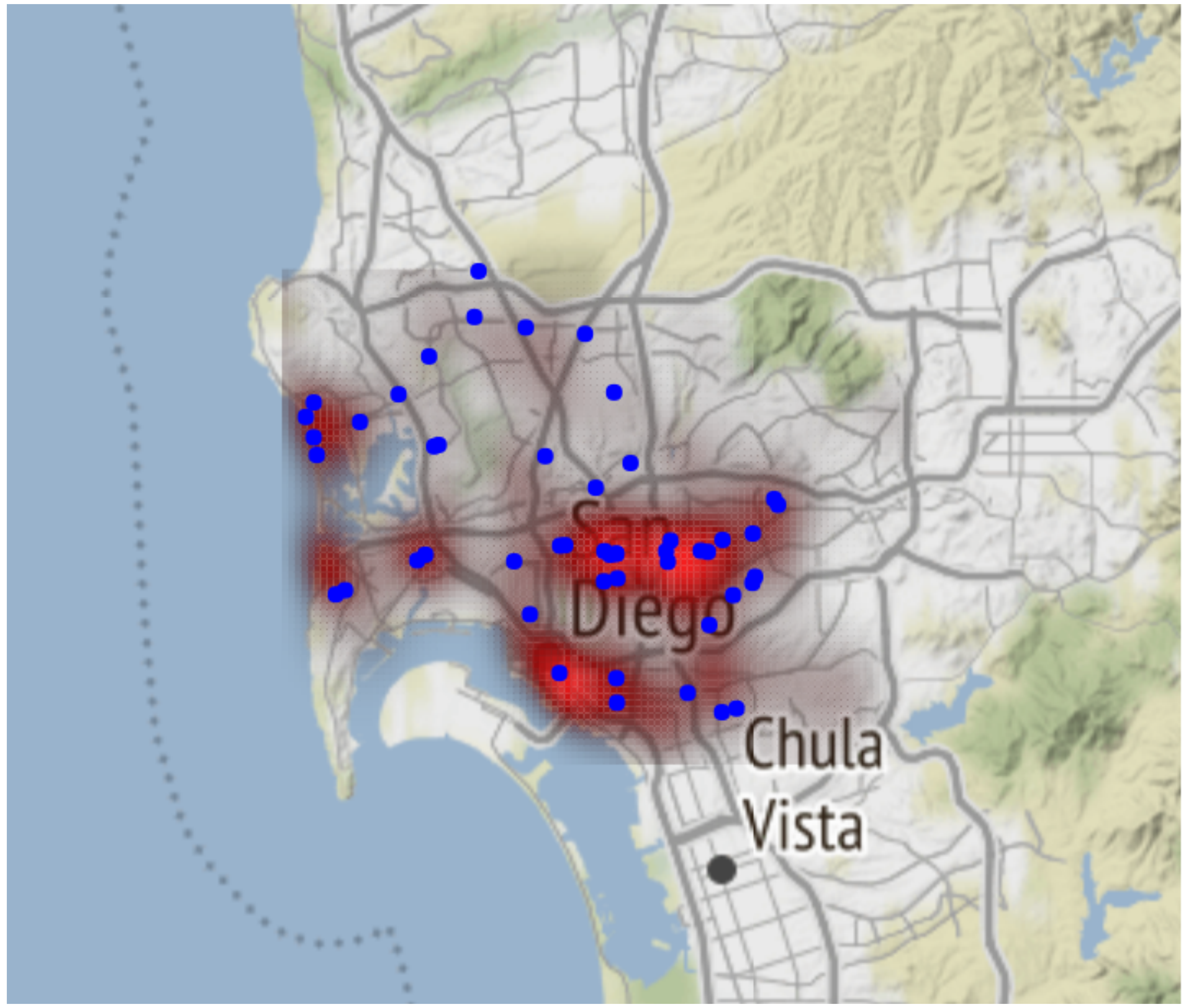


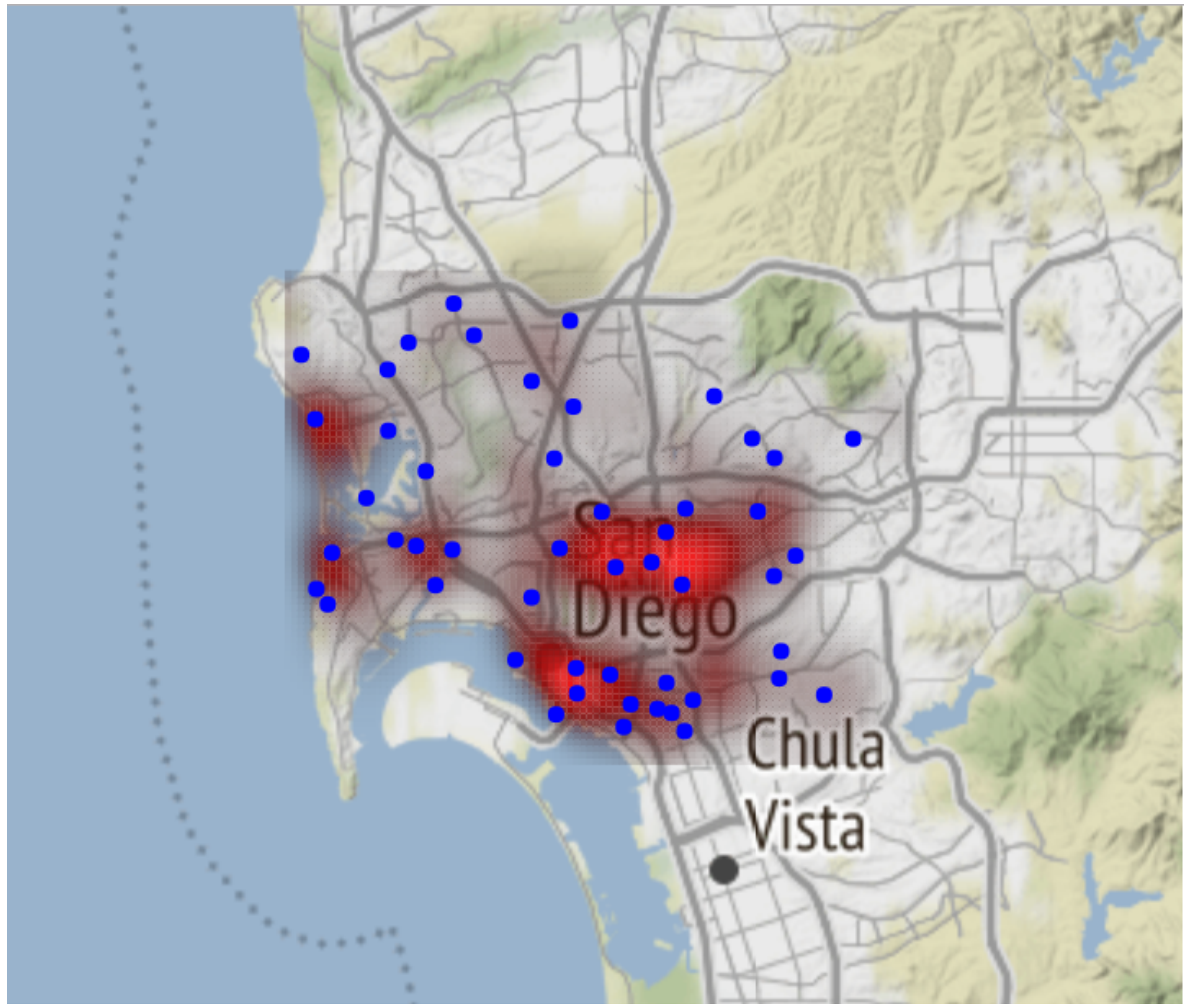




https://github.com/COGS108/FinalProjects-Wi21/blob/main/FinalProject_group084.ipynb

General question: Why isn't police response time always the same?

Data Science question: Where should police cars be stationed, accounting for crime levels and time of day, to make police response times equitable throughout San Diego?




Descriptive

Exploratory

Predictive

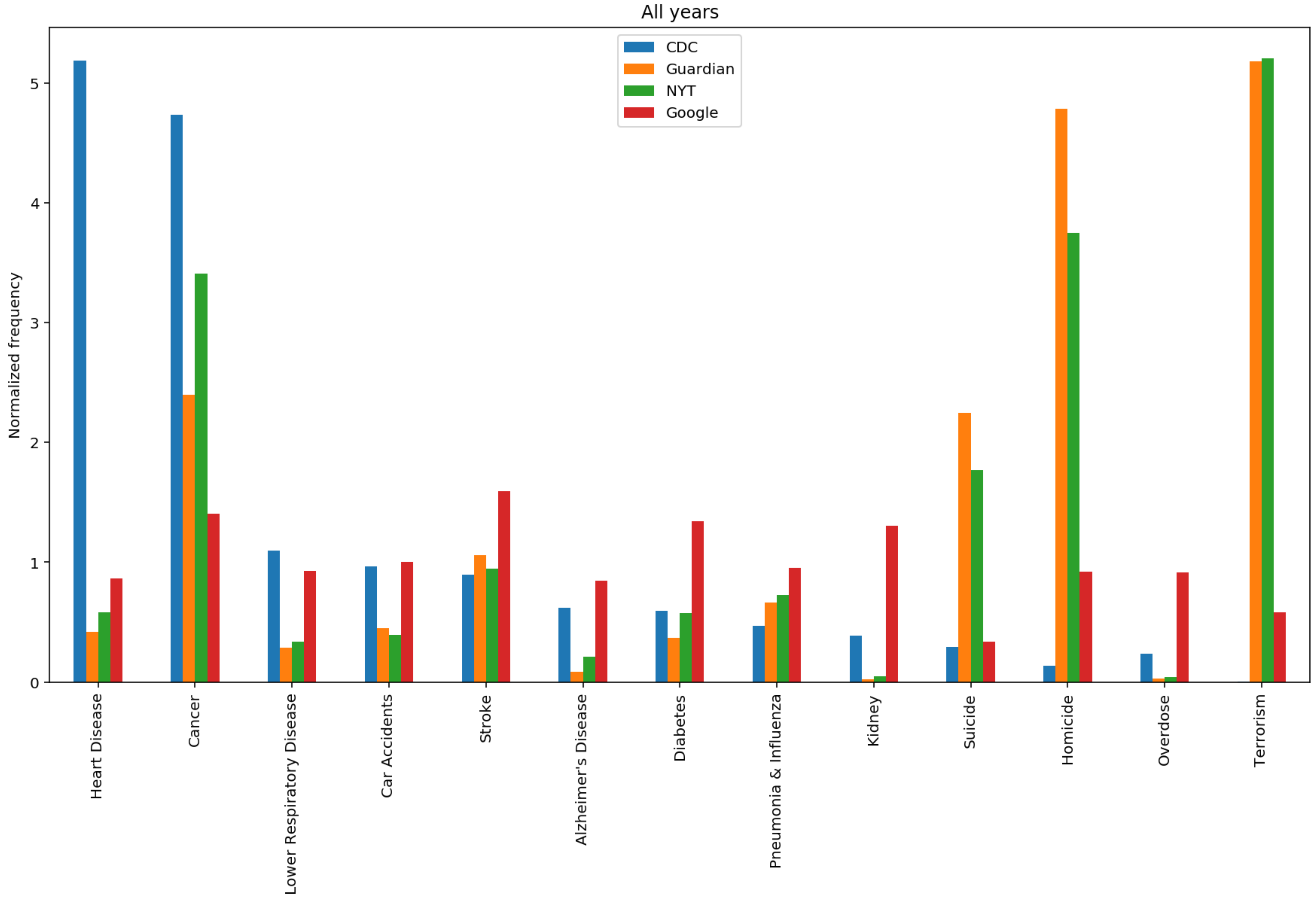
Inferential

Geospatial Analysis

https://github.com/COGS108/FinalProjects-Sp17/blob/master/029-FinalProject.ipynb

General question: What gets too much attention in the news?

Data Science Question: Is there a relationship over time between cause of death terms in the *NYT*, The Guardian, and Google trends data relative to data from the CDC?




**Descriptive**

**Exploratory**

**Inferential**

Text Analysis

Classic Statistics (parametric & nonparametric)

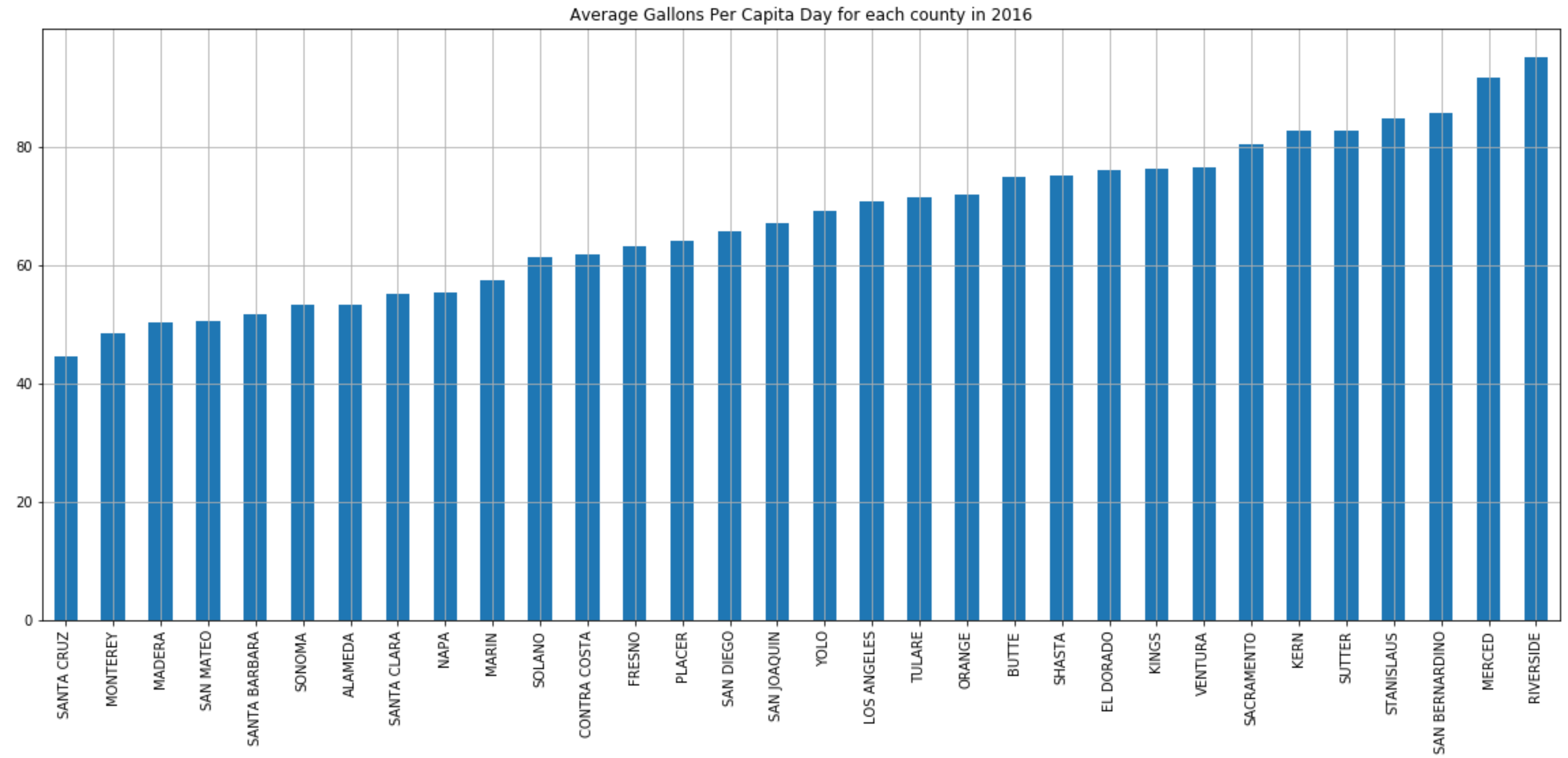
https://github.com/COGS108/FinalProjects-Wi18/blob/master/010-FinalProject.ipynb

*In case of the total drought in California, how many desalination plant projects we need to supply residential use water for population who live in urban areas in California?*

**Descriptive**

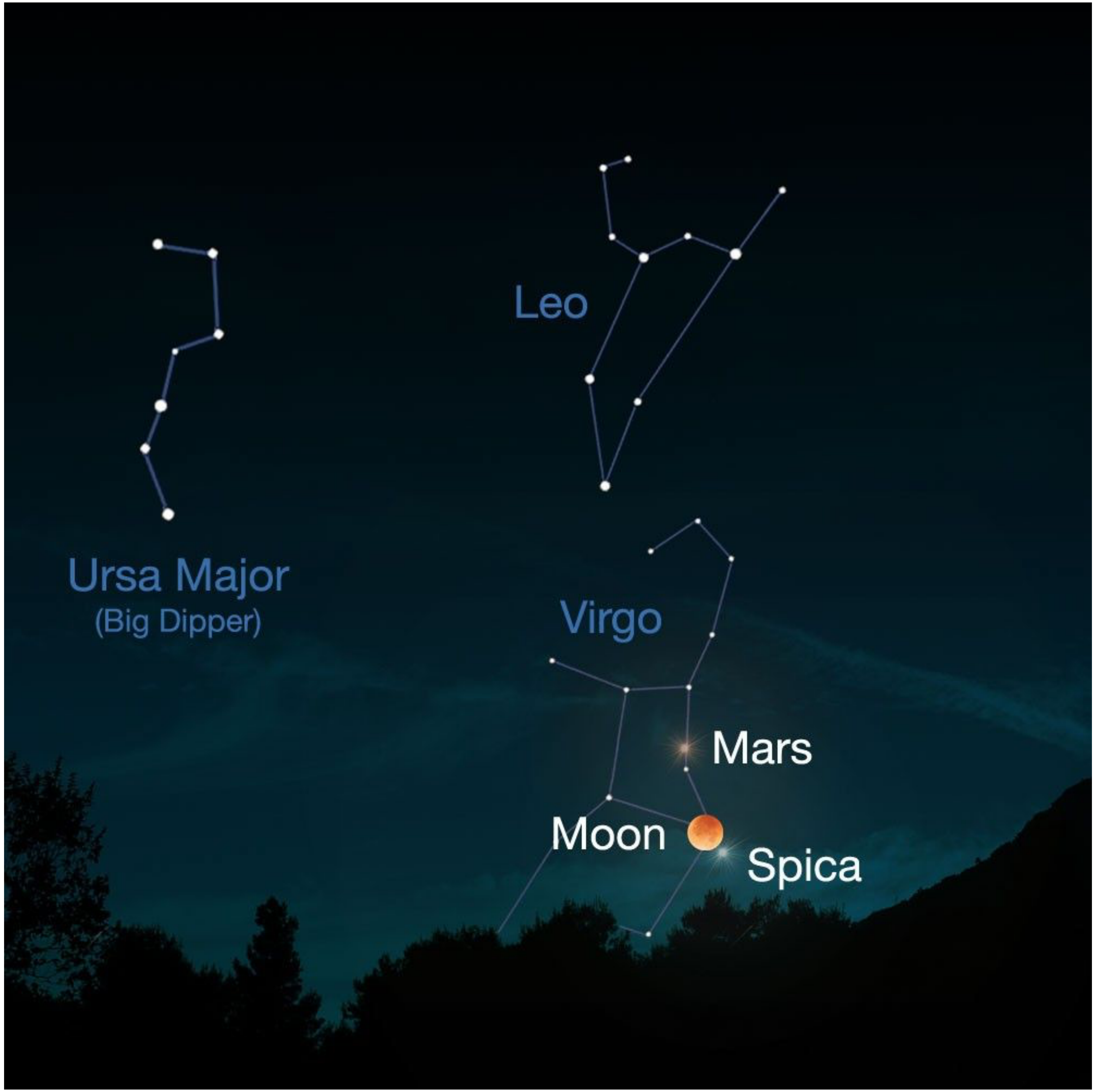
**Exploratory**

**Predictive**




https://github.com/COGS108/FinalProjects-Sp17/blob/master/018-FinalProject.ipynb

# Stars and planets



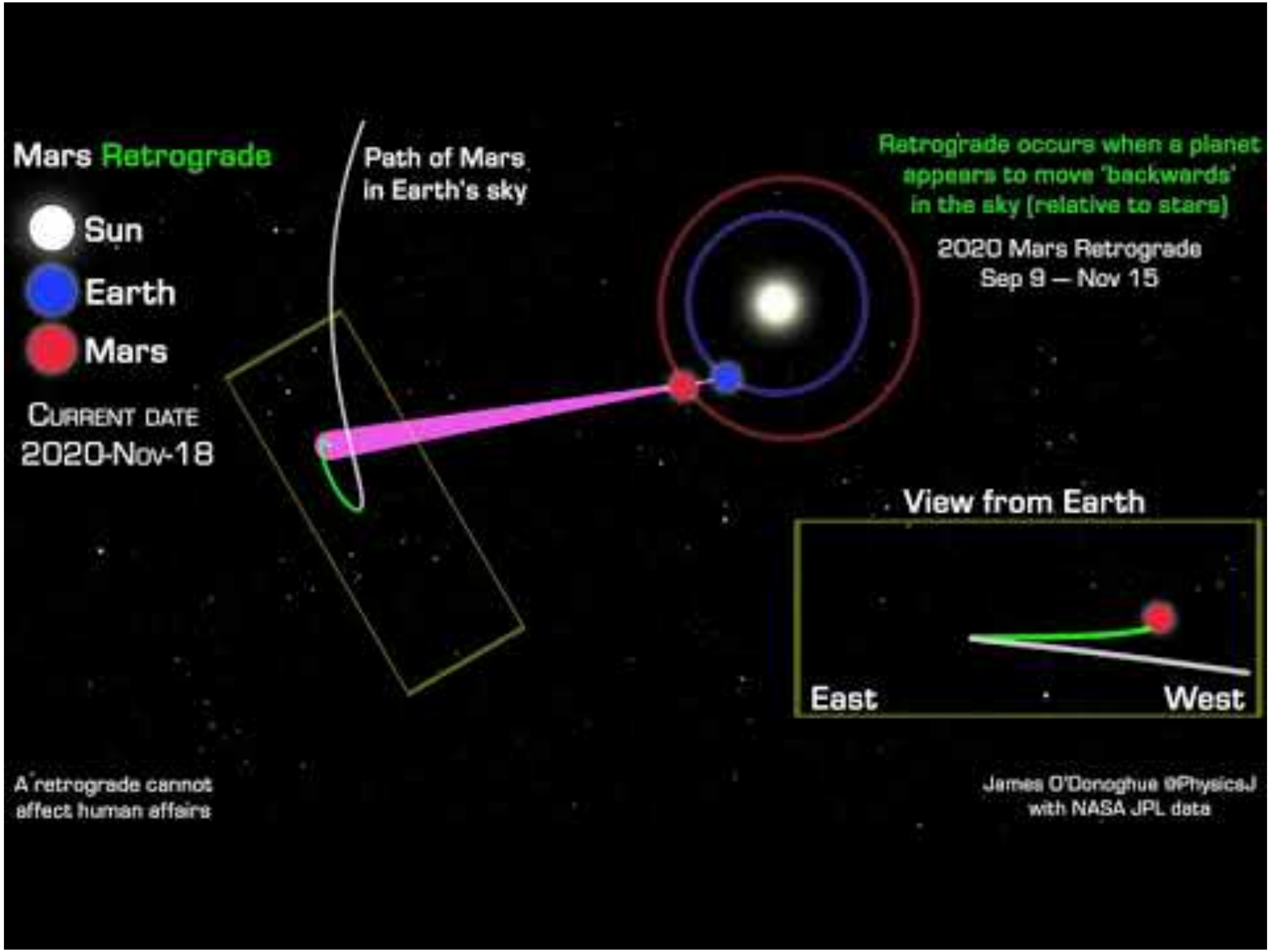
# Mars in retrograde

## Insert Astrology Joke Here

# How can you explain / predict retrograde motion?

## Think-Pair-Share

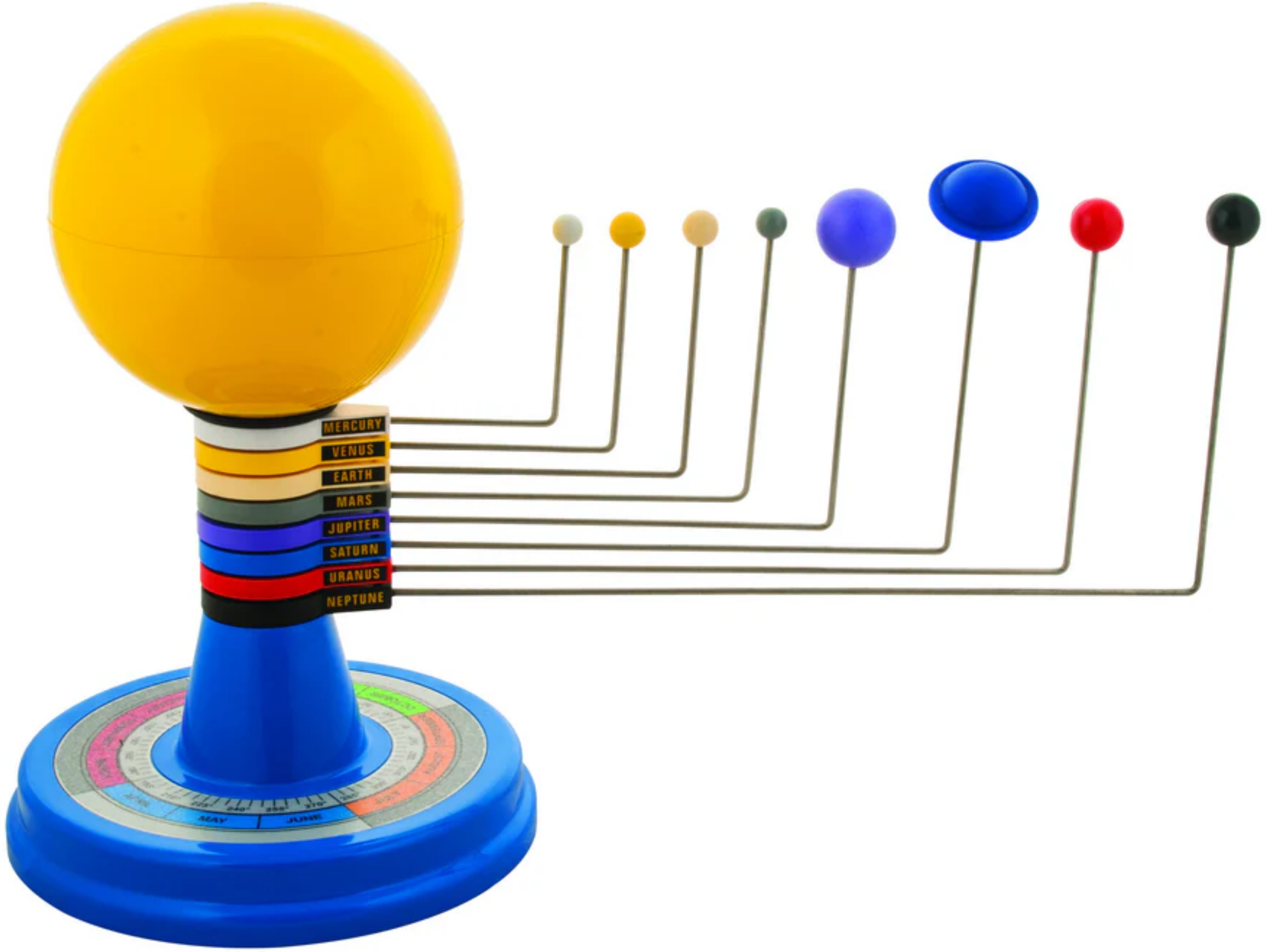
1. Think for yourself 1 min
2. Pair up, introduce yourself, and discuss this topic, 2 to 3 min
3. 3+ hands up and we discuss




https://youtu.be/hOjrPcD6Iuc?si=twTfXeUbz-NVeYZI

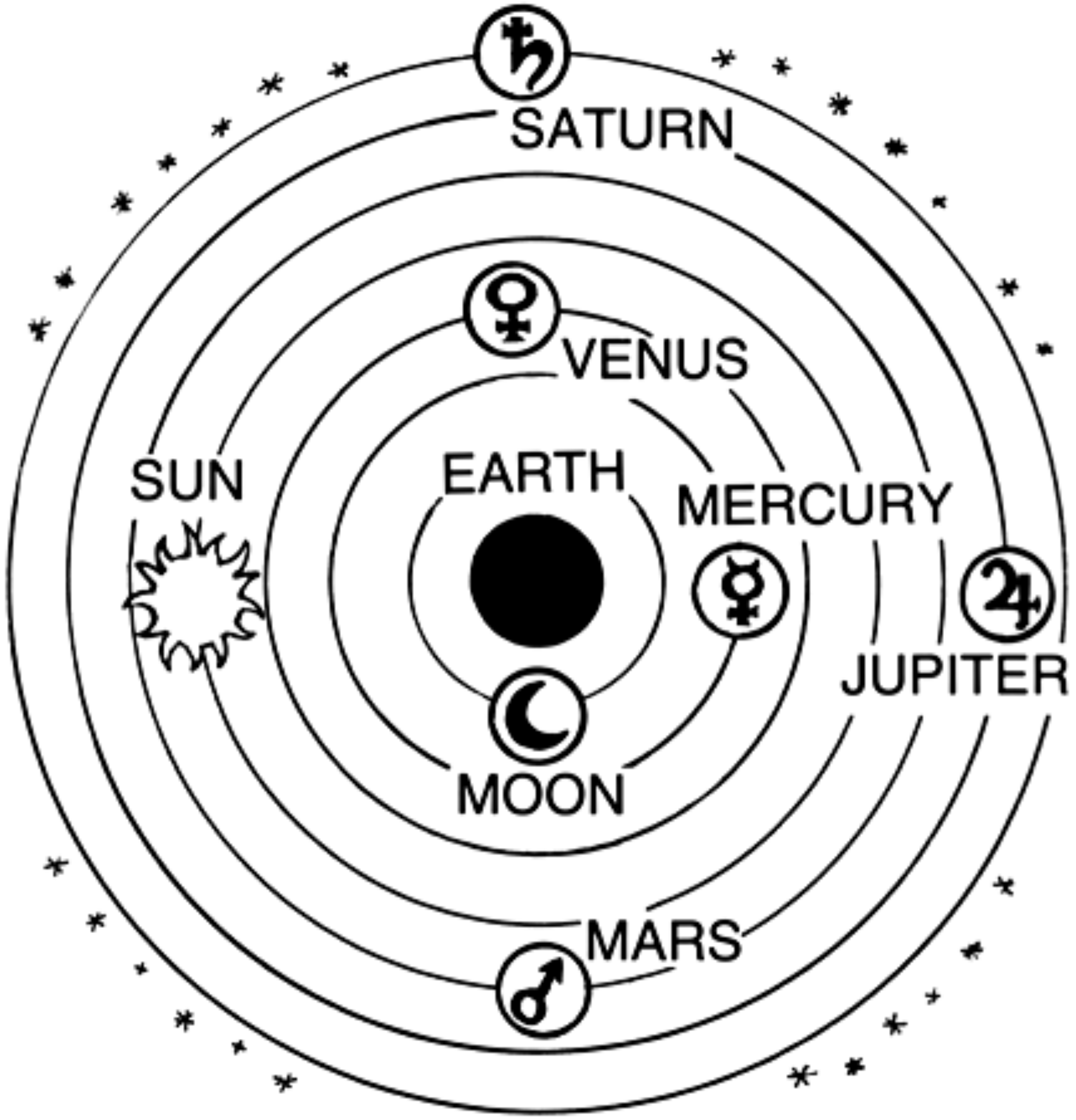
# This is a model!

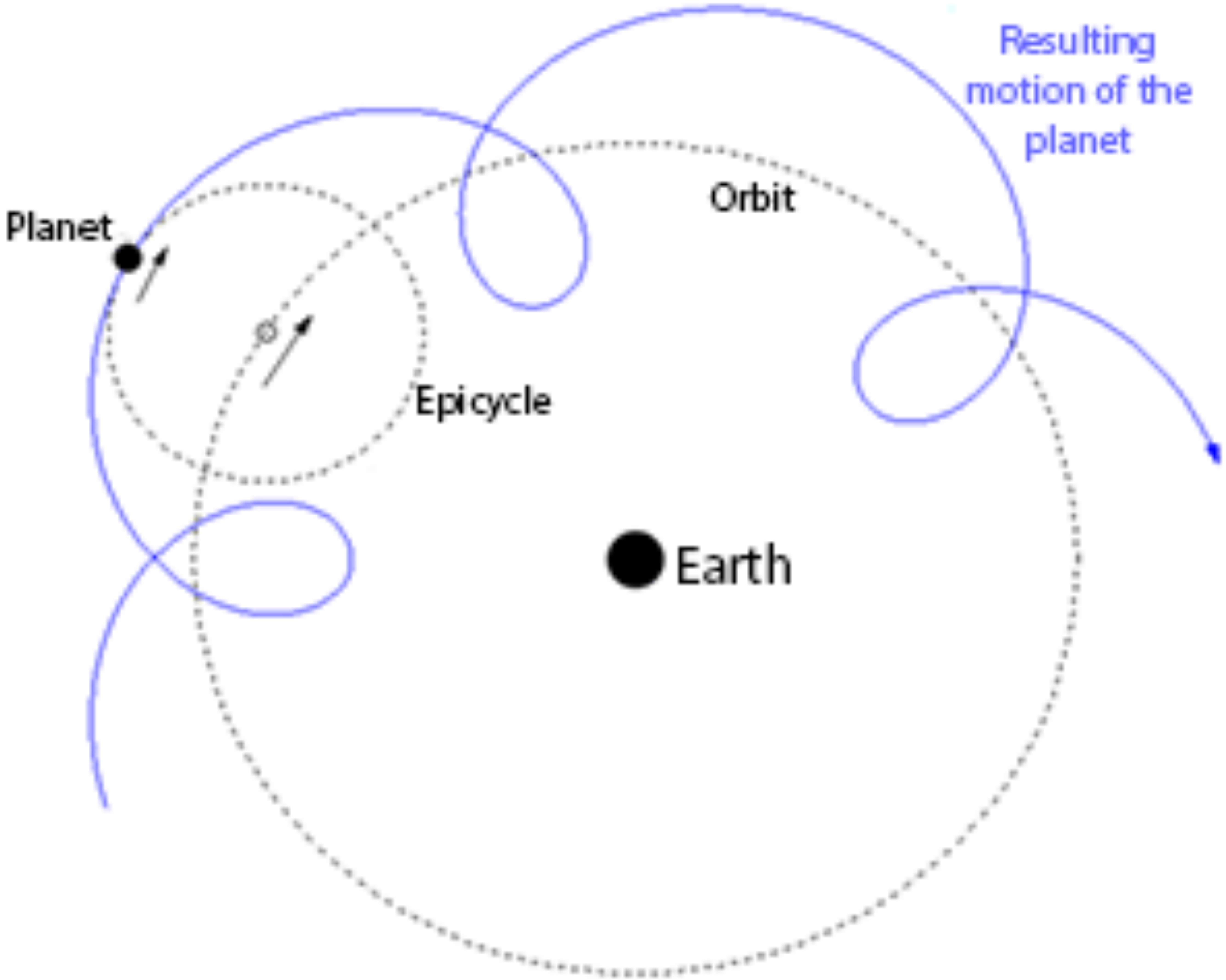
Nicolas Copernicus, 1543




Models (especially math models) help us understand, explain, predict, and analyze

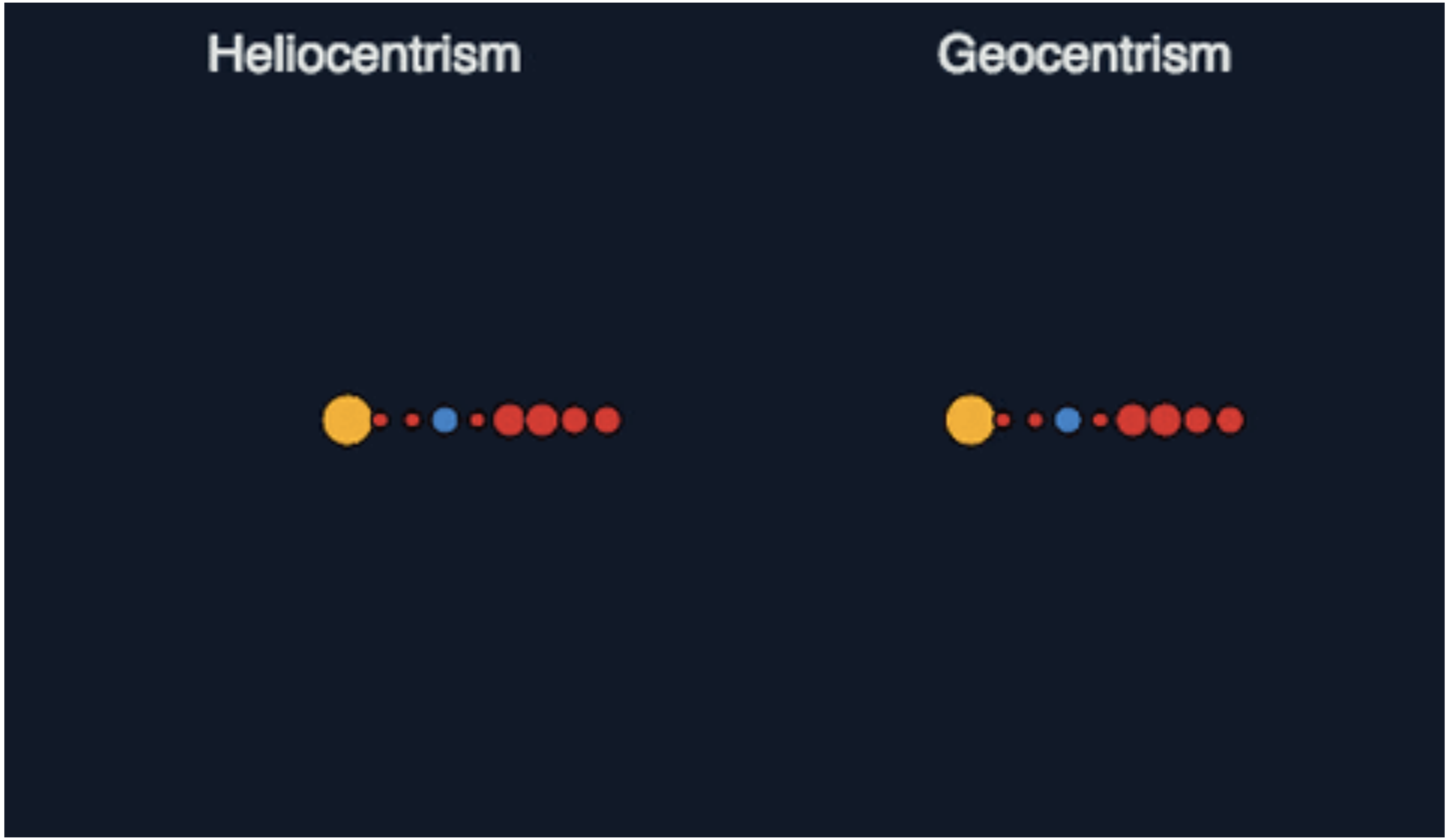
# Before Copernicus



# Both models predict  apparent planetary motion well!

## Moral of the story: it’s easy to screw up data analysis



**Descriptive:** The goal of descriptive analysis is to understand the components of a data set, describe what they are, and explain that description to others who might want to understand the data.

- Problem: Understanding whether users are nice or mean on Youtube
- Data science question: Are the words that people use in their comments more frequently positive words (great, awesome, nice, useful) or negative words (bad, stupid, lame, awful)?
- Type of analysis: Descriptive analysis

To answer this you would calculate <u>statistics</u> about YouTube comments

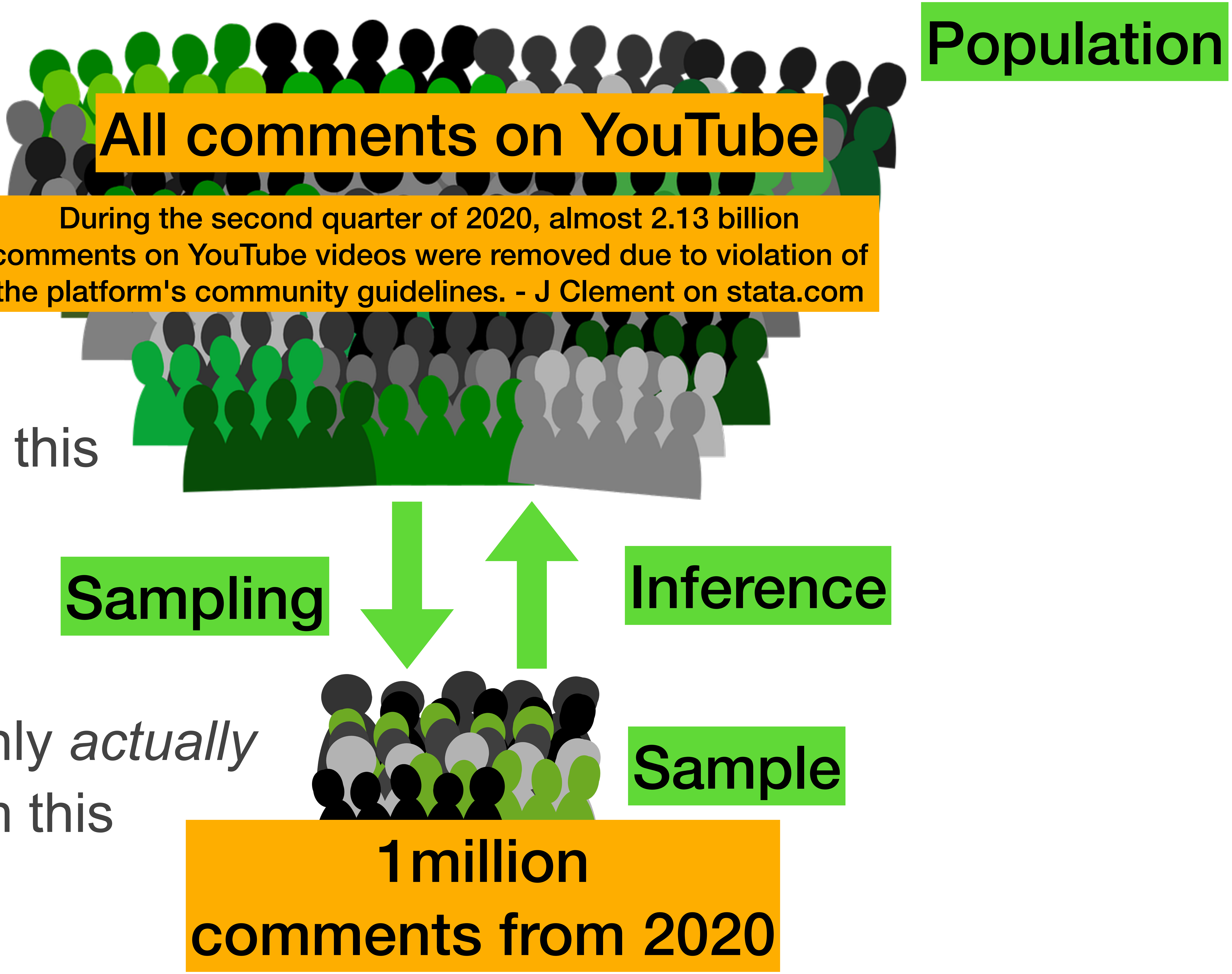
**Statistics:** The science that deals with the collection, classification, analysis, and interpretation of numerical facts or data

# A statistic: a quantity computed from a sample


For our YouTube analysis, we could take a random sample of comments from YouTube and calculate the following statistic: *the number of positive and the number of negative words in each review*.

We want to learn something about this

....but we can only *actually* collect data from this



# Best sampling practices:

- Always think about what your population is
- Collect data from a sample that is representative of your population
- If you have no choice but to work with a dataset that is not collected randomly and is biased, be careful not to generalize your results to the entire population


You'd want to be sure you sample randomly across *all* YouTube comments, making sure not to get more comments from one genre over another, or one location over another, etc.

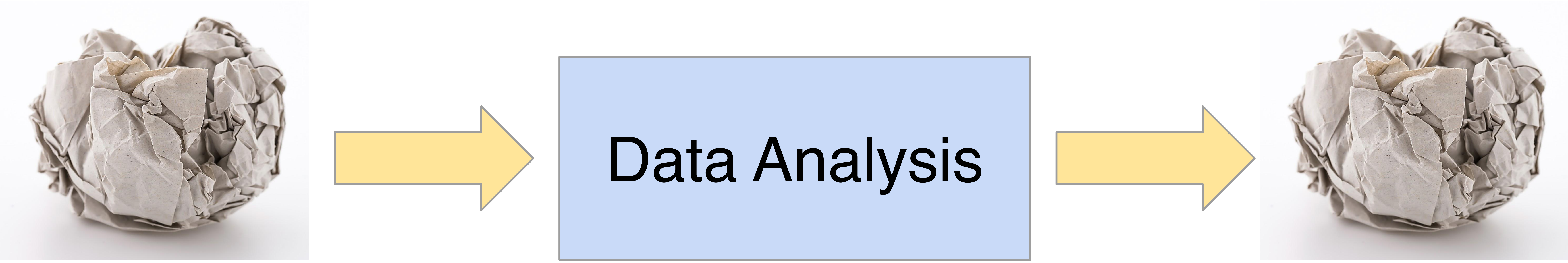
Examples of bad sampling:

- Surveying subscribers of a gun-related magazine for research on Americans' attitudes toward owning guns
- Randomly sampling Facebook users for what TV shows people like


To understand *all* YouTube comments, you wouldn't just want to sample from one YouTube channel, or videos in a single language.

At the <u>beginning</u> of a project determine whether or not the data you have will actually help answer the question

# Garbage In. Garbage Out.



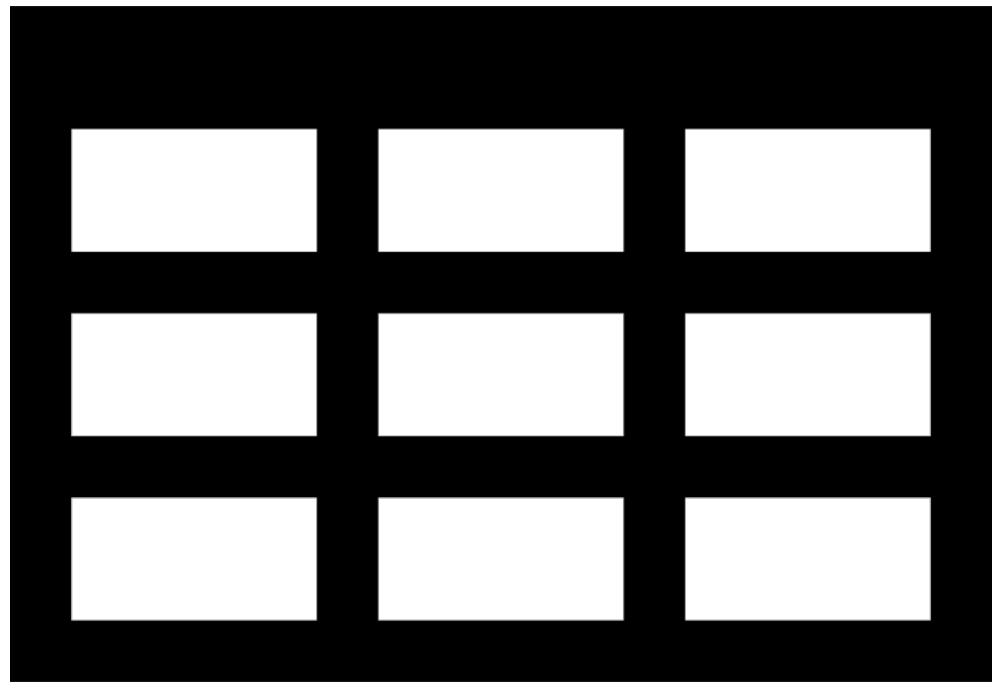
https://forms.gle/v9bn5kCUVLW5Robc6

For the survey data I collected from you all, which of the following best describes the population I could generalize findings back to.

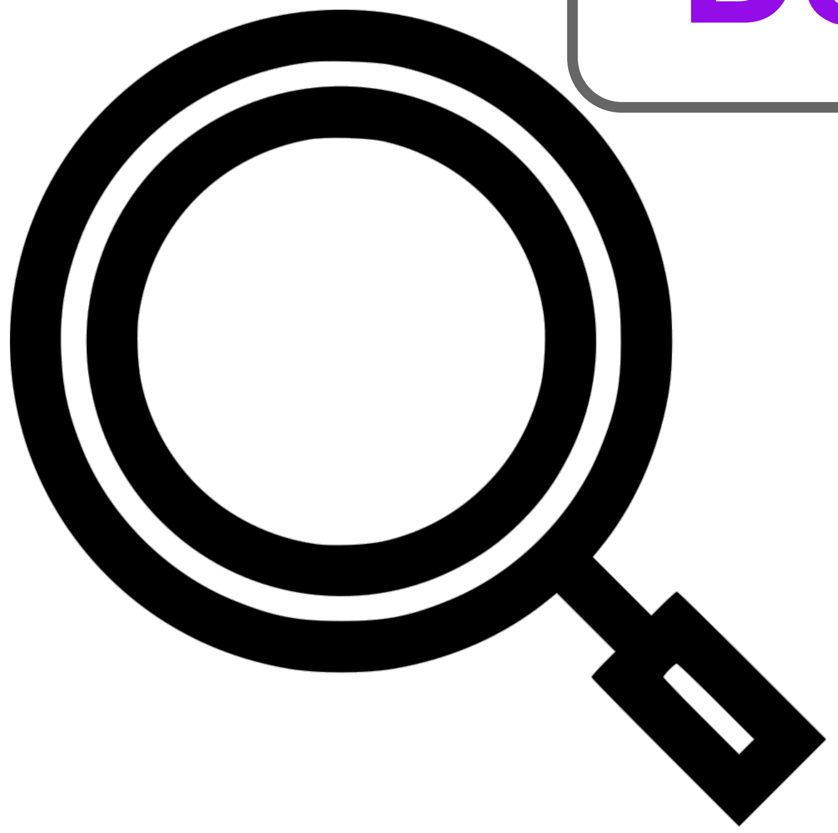
A B C D E

Descriptive

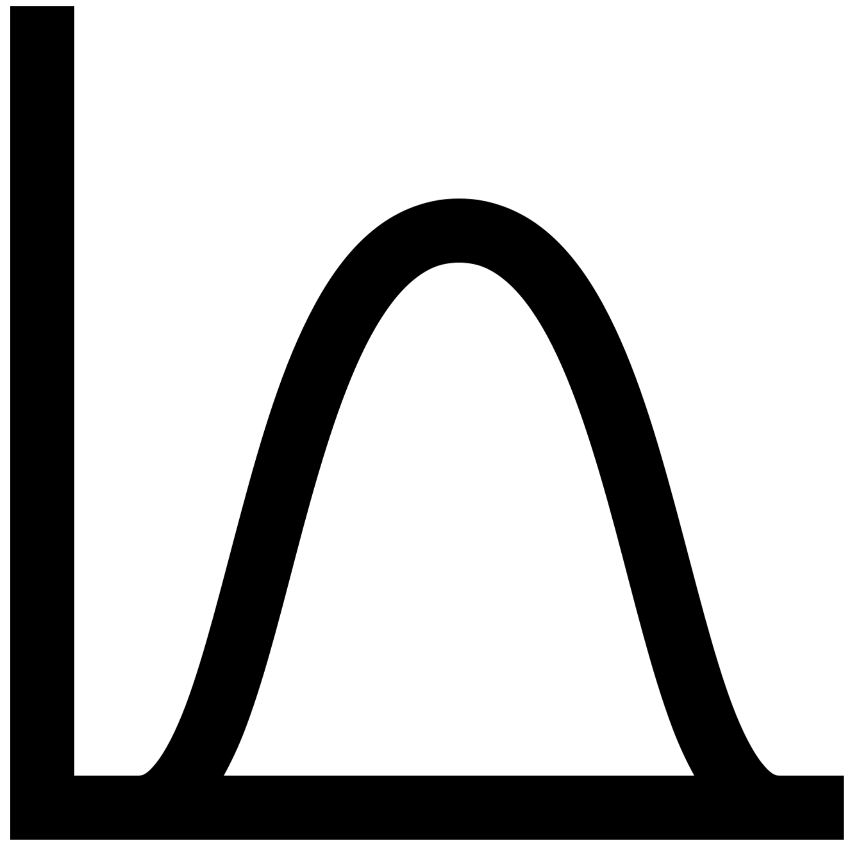
# Descriptive Analysis



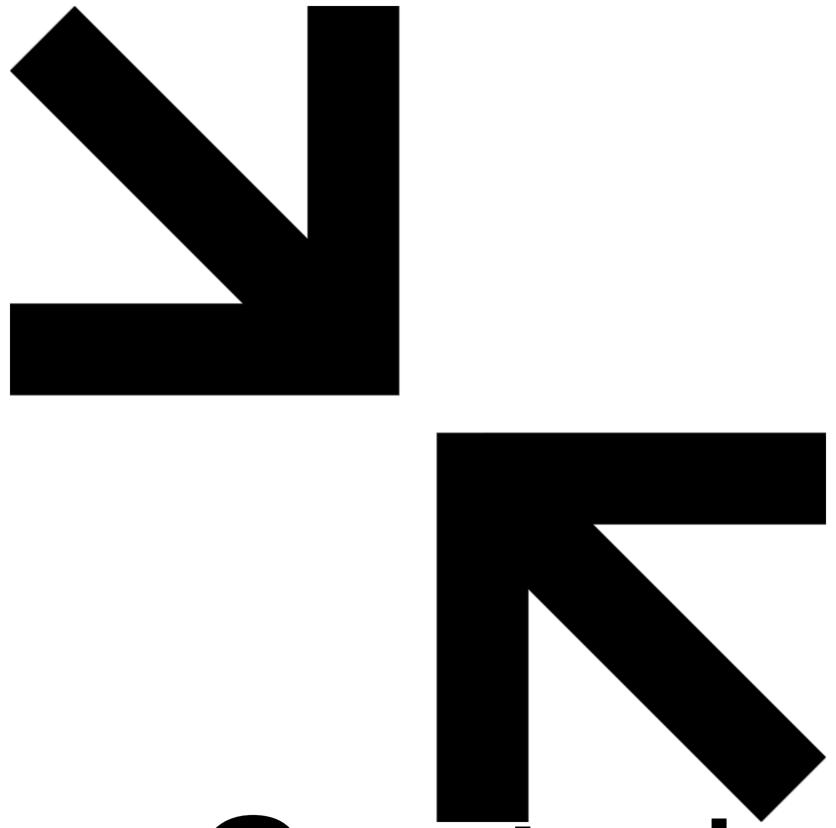
Size



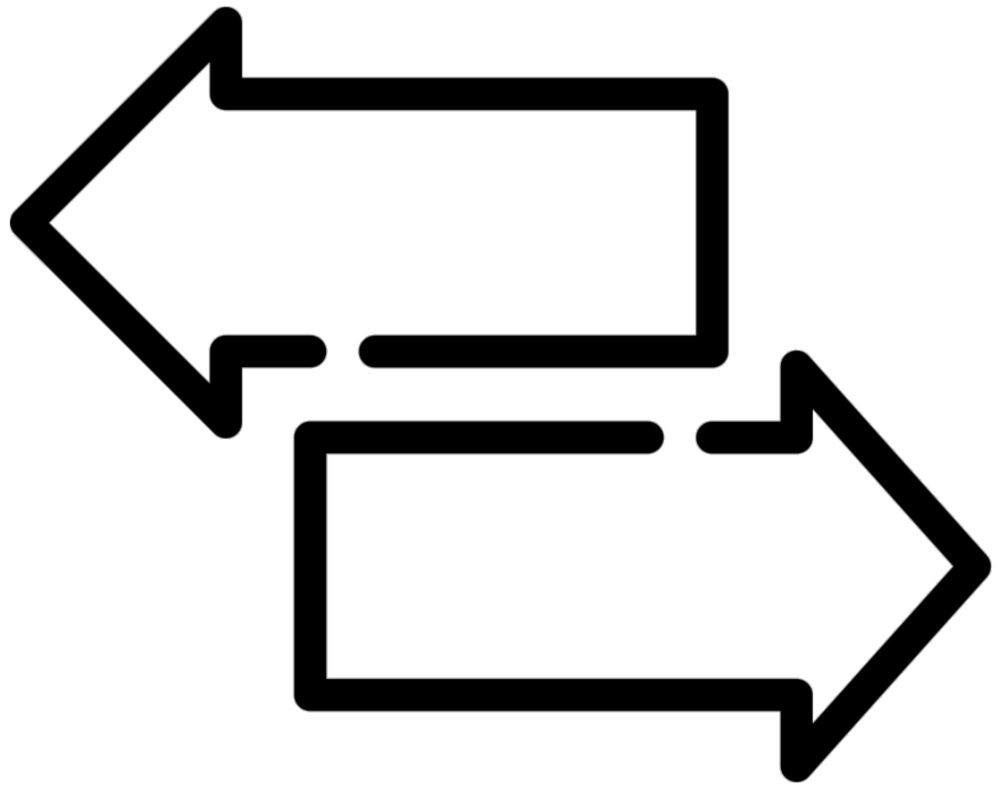
Missingness



Shape

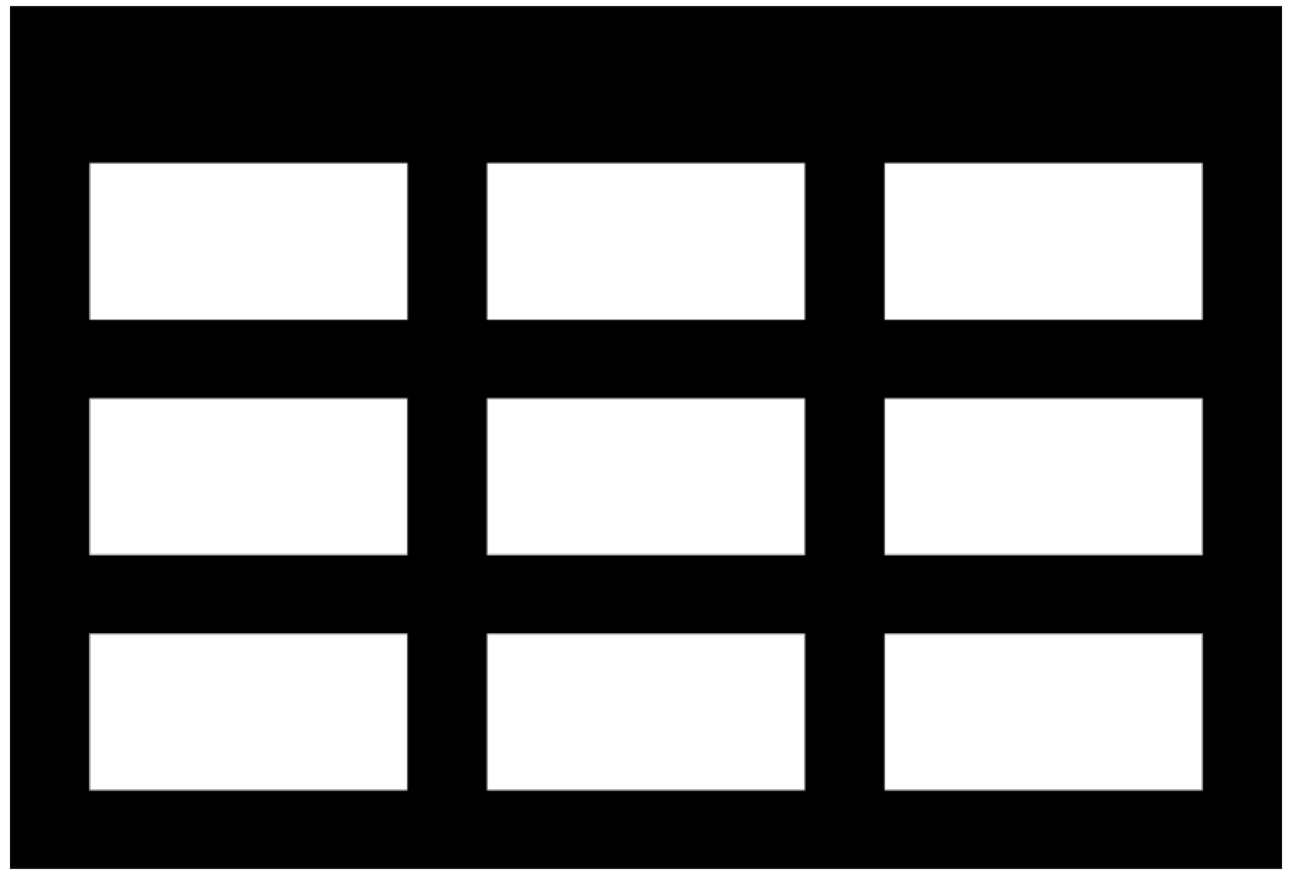


Central Tendency



Variability

Descriptive



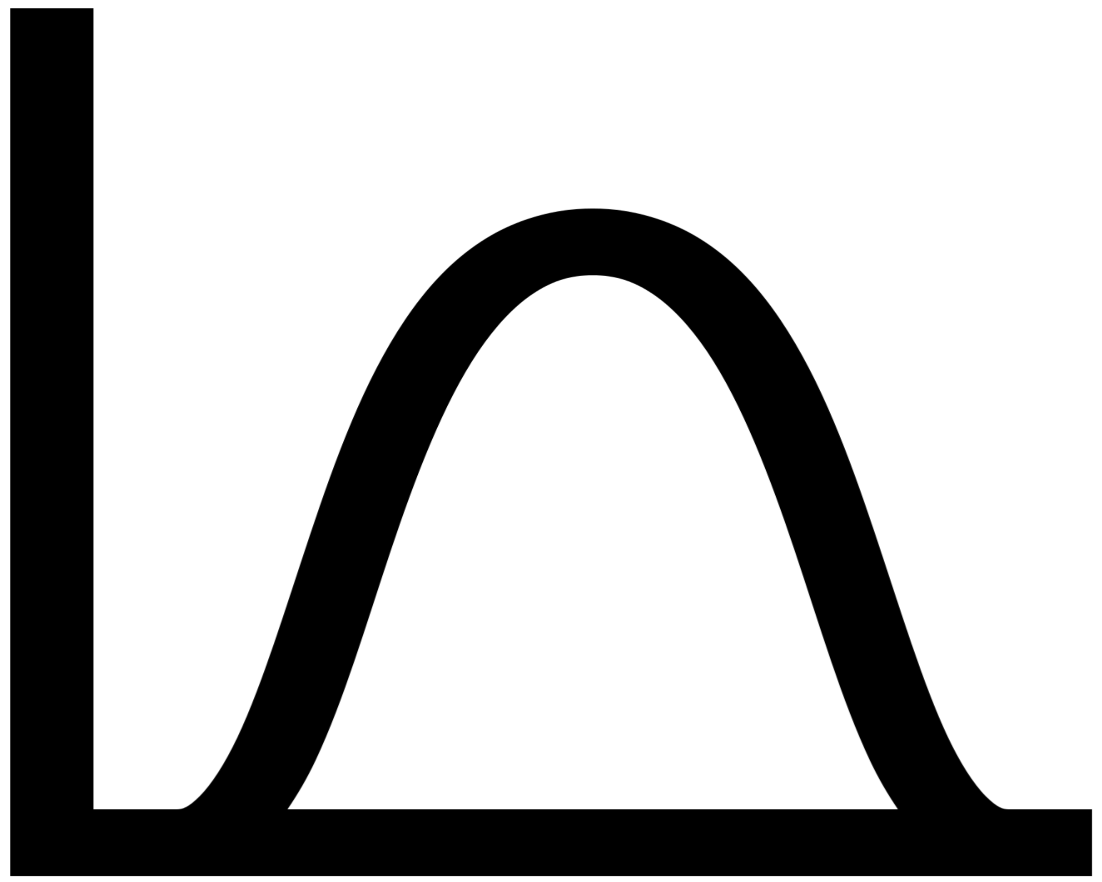
## Size

How many observations (rows) and variables (columns) you have is an important first step. You should always be aware of the size of your dataset

Descriptive

# Missingness

It's critical to know how many observations have missing data for variables of interest in your data. Knowing *why* their missing is also important.



## Shape

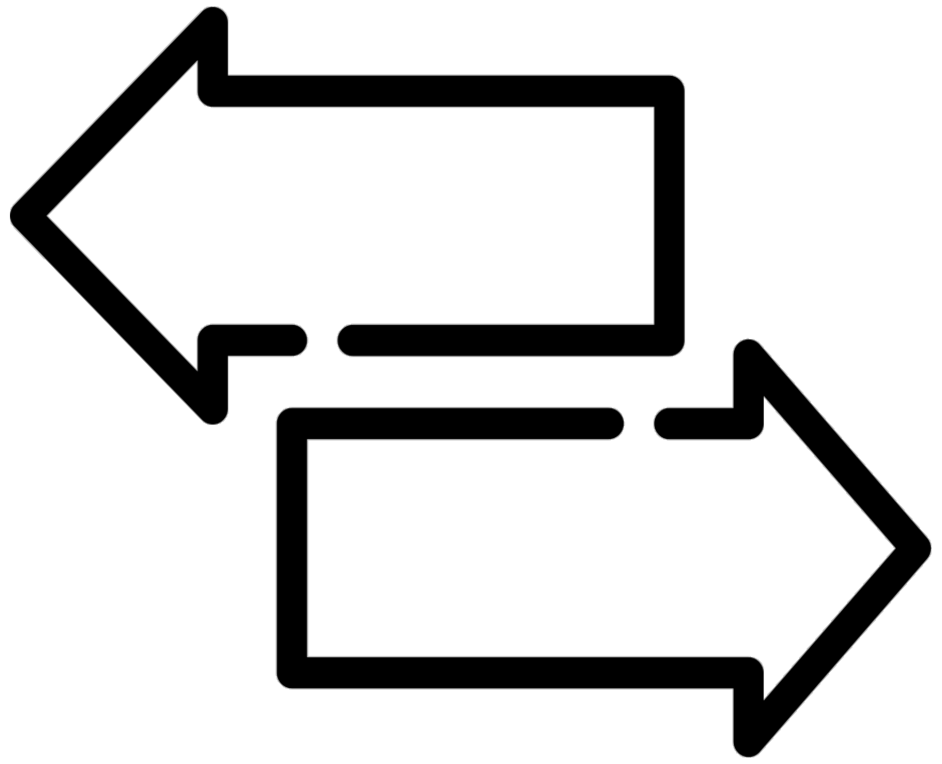
It’s critical to know the distribution of the variables in your dataset. Certain statistical approaches can only be used with certain distributions.

**Descriptive**

## Central Tendency

Knowing the mean, median, and/or mode can help you get an idea of what a typical value is for your variable(s) of interest

Descriptive



# Variability

The central tendency tells you part of the story. The variability in the values in your observation helps fill in the rest.

# Descriptive

Descriptive Analyses are often included as “Table 1” in academic publications

Table 1. Baseline Characteristics of the Patients.*

| Characteristic | Ranibizumab Monthly (N=301) | Bevacizumab Monthly (N=286) | Ranibizumab as Needed (N=298) | Bevacizumab as Needed (N=300) |
|---|---|---|---|---|
| Age — no. (%) | | | | |
| 50–59 yr | 2 (0.7) | 1 (0.3) | 6 (2.0) | 2 (0.7) |
| 60–69 yr | 33 (11.0) | 28 (9.8) | 31 (10.4) | 34 (11.3) |
| 70–79 yr | 102 (33.9) | 84 (29.4) | 115 (38.6) | 103 (34.3) |
| 80–89 yr | 142 (47.2) | 150 (52.4) | 126 (42.3) | 142 (47.3) |
| ≥90 yr | 22 (7.3) | 23 (8.0) | 20 (6.7) | 19 (6.3) |
| Mean — yr | 79.2±7.4 | 80.1±7.3 | 78.4±7.8 | 79.3±7.6 |
| Sex — no. (%) | | | | |
| Female | 183 (60.8) | 180 (62.9) | 185 (62.1) | 184 (61.3) |
| Male | 118 (39.2) | 106 (37.1) | 113 (37.9) | 116 (38.7) |
| Race — no. (%)† | | | | |
| White | 297 (98.7) | 281 (98.3) | 296 (99.3) | 294 (98.0) |
| Other | 4 (1.3) | 5 (1.7) | 2 (0.7) | 6 (2.0) |
| History of myocardial infarction — no. (%) | 34 (11.3) | 40 (14.0) | 30 (10.1) | 36 (12.0) |
| History of stroke — no. (%) | 14 (4.7) | 18 (6.3) | 22 (7.4) | 16 (5.3) |
| History of transient ischemic attack — no. (%) | 12 (4.0) | 25 (8.7) | 12 (4.0) | 19 (6.3) |
| Blood pressure — mm Hg | | | | |
| Systolic | 134±18 | 135±19 | 136±17 | 135±17 |
| Diastolic | 75±10 | 75±10 | 76±9 | 75±10 |
| Visual-acuity score and Snellen equivalent | | | | |
| 68–82 letters, 20/25–40 — no. (%) | 111 (36.9) | 94 (32.9) | 116 (38.9) | 103 (34.3) |
| 53–67 letters, 20/50–80 — no. (%) | 98 (32.6) | 118 (41.3) | 108 (36.2) | 119 (39.7) |
| 38–52 letters, 20/100–160 — no. (%) | 67 (22.3) | 53 (18.5) | 58 (19.5) | 58 (19.3) |
| 23–37 letters, 20/200–320 — no. (%) | 25 (8.3) | 21 (7.3) | 16 (5.4) | 20 (6.7) |
| Mean score | 60.1±14.3 | 60.2±13.1 | 61.5±13.2 | 60.4±13.4 |
| Total thickness at fovea — μm‡ | 458±184 | 463±196 | 458±193 | 461±175 |
| Retinal thickness plus subfoveal-fluid thickness at fovea — μm | 251±122 | 254±121 | 247±122 | 252±115 |
| Foveal center involvement — no. (%) | | | | |
| Choroidal neovascularization | 176 (58.5) | 153 (53.5) | 176 (59.1) | 183 (61.0) |
| Fluid | 85 (28.2) | 81 (28.3) | 77 (25.8) | 72 (24.0) |
| Hemorrhage | 20 (6.6) | 24 (8.4) | 24 (8.1) | 25 (8.3) |
| Other | 18 (6.0) | 20 (7.0) | 15 (5.0) | 18 (6.0) |
| No choroidal neovascularization or not possible to grade | 2 (0.7) | 8 (2.8) | 6 (2.0) | 2 (0.7) |

* Plus–minus values are means ±SD.
† Race was self-reported.
‡ Total thickness at the fovea includes the retina, subretinal fluid, choroidal neovascularization, and retinal pigment epithelial elevation.

# Descriptive

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† Race was self-reported.
‡ Total thickness at the fovea includes the retina, subretinal fluid, choroidal neovascularization, and retinal pigment epithelial elevation.

Size

Shape

Central tendency

variability

Zooming in on this we see variables stratified by Age, Sex, and Race

# Stratification changes results

Sample: 400 patients with index vertebral fractures

…looks like vertebroplasty was *way* worse for patients!

| Vertebroplasty | Conservative care | Relative risk (95% confidence interval) |
|---|---|---|
| 30/200 (15%) | 15/200 (7.5%) | 2.0 (1.1–3.6) |

subsequent fractures

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3503514/

# But wait...at time of initial fracture...

| | Vertebroplasty N = 200 | Conservative care N = 200 |
|---|---|---|
| Age, y, mean ± SD | 78.2 ± 4.1 | 79.0 ± 5.2 |
| Weight, kg, mean ± SD | 54.4 ± 2.3 | 53.9 ± 2.1 |
| Smoking status, No. (%) | 110 (55) | 16 (8) |

Age and weight are similar between groups. **Smoking Status** differs vastly.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3503514/

# So...let's stratify those results real quick

| Smoke | | | No smoke | | |
|---|---|---|---|---|---|
| Vertebroplasty | Conservative | RR (95% confidence interval) | Vertebroplasty | Conservative | RR (95% confidence interval) |
| 23/110 (21%) | 3/16 (19%) | 1.1 (0.4, 3.3) | 7/90 (8%) | 12/184(7%) | 1.2 (0.5, 2.9) |

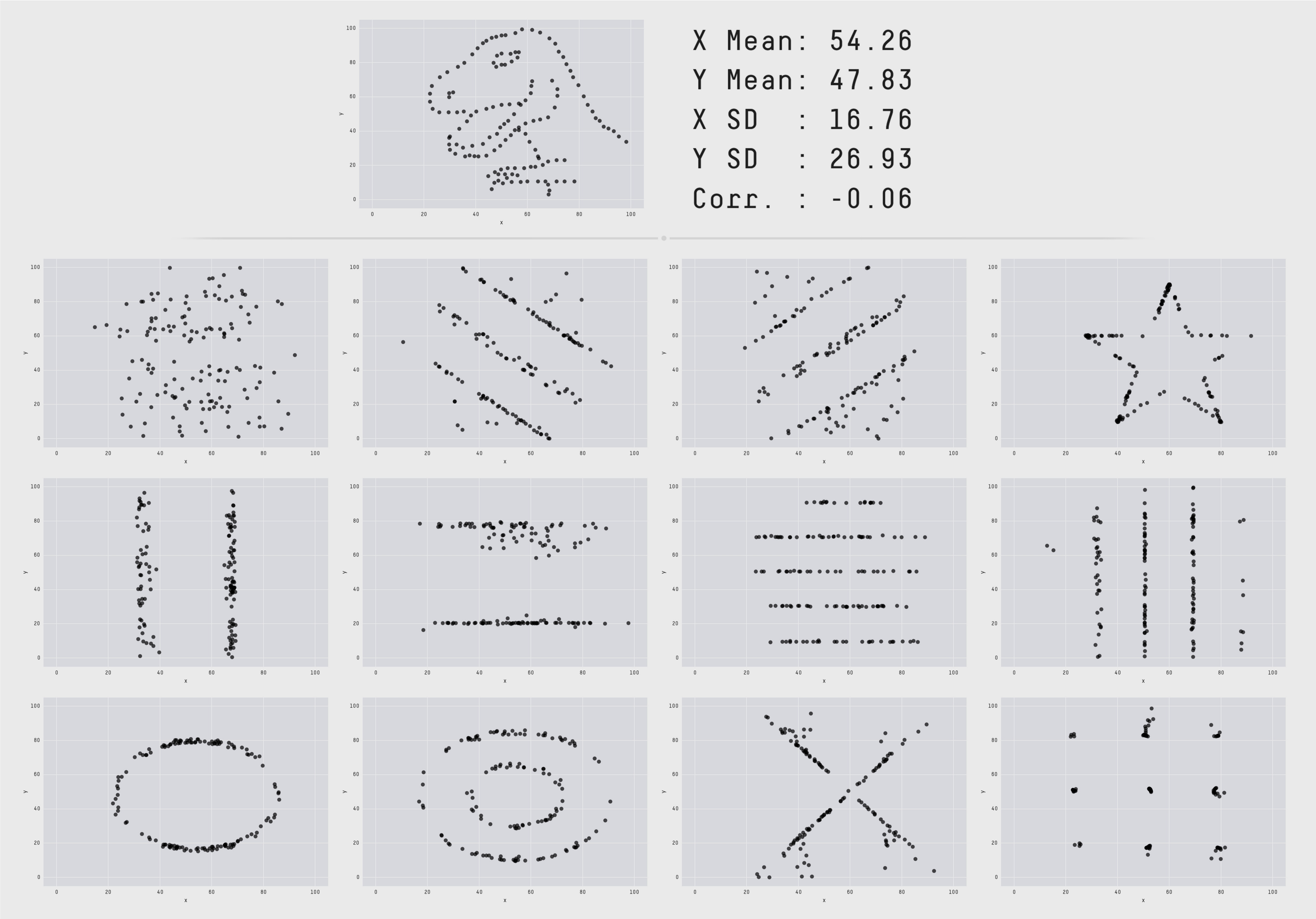
Risk of re-fracture is now similar within group

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3503514/

# Descriptive Statistics & Summary

“*We must suppress some of the truth to communicate the truth... In short, the techniques of descriptive statistics are designed to match the salient features of the data set to human cognitive abilities.*”

-I.J. Good (1983)




https://www.autodesk.com/research/publications/same-stats-different-graphs

A modern version of Anscombe’s Quartet